

Assignment 4

Due: April 15, 2026 at 11:59 pm (Central Time)

Hierarchical clustering is sometimes used to generate K clusters, $K > 1$, by taking the clusters at the K^{th} level of the dendrogram (Root is at level 1). By looking at the clusters produced in this way, we can evaluate the behavior of hierarchical clustering and compare it with the behavior of K-means. Following is a set of one-dimensional points: {6, 12, 18, 24, 30, 42, 48}.

- (a) Please conduct Complete Linkage agglomerative hierarchical clustering, and
 - (30%) compute the distance/proximity matrix after each merging step.
 - (10%) show the result by drawing a dendrogram. The dendrogram should clearly show the order in which the points are merged.
 - (10%) What are the two clusters produced by Complete Linkage approach?

- (b)(30%) For each of the following sets of initial centroids, please create two clusters by assigning each point to the nearest centroid, and then calculate the total squared error for each set of two clusters. Please show both the clusters and the total squared error for each set of centroids.
 - {18,45}
 - {15,40}

- (c) (20%) Do both sets of centroids (given in part b) represent stable solutions? i.e., if the K-means algorithm is run on this set of points using the given centroids as the starting centroids, will there be any change in the final clusters returned by the algorithm? Please provide the outcome of each iteration of K-means when being run on each set of starting centroids.